

Software Requirements Specification (SRS)

Project: Vehicle Parking System

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Revision History

Version	Date	Author	Change Summary	Approval
1.0	06-09-2026	Team 10	Initial SRS draft	Pending
1.1	28-09-2026	Team 10	Added PlantUML use-case diagrams	Pending

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for a Vehicle Parking System. It defines functional and non-functional requirements, external interfaces, and verification criteria for use by developers, QA engineers, and the course assessor. The system automates the management of parking slots, vehicle entry/exit, fee calculation, and administrative monitoring.

1.2 Scope

Covers all user interactions at parking entry/exit terminals, real-time slot management, vehicle tracking, fee calculation, receipt generation, and administrative reporting. Excludes payment gateway internal processing beyond API integration, physical barrier hardware firmware, and city-level traffic management systems.

1.3 Audience

Developers, QA Engineers, System Integrators, Maintenance Technicians, and Assessment Evaluators.

1.4 Definitions

Term	Definition
VPS	Vehicle Parking System
LP	License Plate
ALPR	Automatic License Plate Recognition
OTP	One-Time Password
API	Application Programming Interface
UI	User Interface
TLS	Transport Layer Security
RBAC	Role-Based Access Control
SRS	Software Requirements Specification
RTM	Requirements Traceability Matrix

2. Overall Description

2.1 Product Perspective

The Vehicle Parking System is a self-service software platform that manages parking facilities. It interacts with entry/exit barrier hardware, a central database for slot tracking, a payment gateway for fee collection, and an admin dashboard for monitoring. The system operates in real time and provides both customer-facing terminals and a web-based admin console.

2.2 Major Product Functions

- Vehicle entry registration (license plate capture, ticket generation)
- Real-time parking slot availability display
- Slot assignment and reservation
- Vehicle exit processing and fee calculation
- Receipt and ticket printing
- Payment processing (cash / card / digital wallet)
- Admin dashboard: reports, occupancy overview, revenue tracking
- Alerts for full capacity, unauthorized vehicles, and system faults

- Offline queuing for transactions during network outages

2.3 User Roles and Characteristics

Role	Description
Driver / Customer	General public; expects quick entry/exit, clear instructions, and accurate billing
Parking Attendant	On-site staff; handles manual overrides and assists customers
System Administrator	Manages slot configuration, user roles, system settings, and reports
Maintenance Technician	Diagnoses hardware and software faults; accesses diagnostic logs

2.4 Operating Environment

Web-based admin portal (modern browsers), embedded terminal software at entry/exit points, networked via secure WAN/LAN, power backed by UPS, operating in outdoor environmental conditions (temperature/humidity rated hardware).

2.5 Constraints

- PCI-DSS compliance for payment card data handling
- ALPR camera integration is vendor-specific
- Hardware barrier APIs vary by vendor and must be abstracted
- System must operate in degraded (offline) mode during network outages
- WCAG 2.1 AA accessibility compliance for the customer-facing UI

3. External Interface Requirements

3.1 User Interfaces

- **Entry Terminal:** Touch-screen kiosk with license plate input, ticket dispensing, slot availability display. High-contrast mode and audio prompts for accessibility.
- **Exit Terminal:** Touch-screen kiosk for ticket scan/entry, payment processing, and receipt printing.
- **Admin Web Portal:** Responsive web dashboard for slot management, occupancy reports, revenue analytics, and user management.

3.2 Hardware Interfaces

- ALPR camera module (vendor API integration)
- Entry/exit barrier gate controller (vendor-specific serial/network interface)
- Ticket printer (ESC/POS protocol)
- Receipt printer
- Card payment terminal (PCI-compliant POS device)

- Status LEDs and operator panel at entry/exit booths

3.3 Software Interfaces

- **Payment Gateway API:** JSON over TLS for card/digital payment authorisation and settlement
- **ALPR Service API:** Image-to-text license plate recognition results
- **Monitoring & Logging API:** Health checks, uptime alerts, and audit log shipping
- **SMS/Email Notification Service:** Entry/exit confirmations and reservation reminders

3.4 Communications

- TLS 1.2+ enforced for all network communication
- Mutual TLS for payment gateway connections
- Retry/backoff strategy for failed API calls
- Offline transaction queuing with secure signing when network is unavailable

4. System Features

The system contains **15 functional requirements**, **5 non-functional requirements**, **2 security objectives**, and **5 security requirements** as required.

4.1 Vehicle Entry & Ticket Generation

Description: When a vehicle arrives, the system captures the license plate, assigns an available slot, generates a unique ticket, and opens the barrier.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
VPS-F-001	The system shall capture the vehicle license plate at entry using ALPR or manual input.	Functional	High	Operations	AC: License plate stored with timestamp on entry. Test: TC-ENT-01	Requires ALPR camera or manual fallback

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
VPS-F-002	The system shall assign an available parking slot to the vehicle upon entry and mark it as occupied.	Functional	High	Operations	AC: Slot status changes to "Occupied"; ticket reflects slot number. Test: TC-ENT-02	Requires real-time slot DB
VPS-F-003	The system shall generate a unique ticket with entry timestamp, slot number, and license plate, and print or display it to the customer.	Functional	High	Customer	AC: Ticket generated within 3 seconds of entry; unique ID confirmed. Test: TC-ENT-03	Depends on VPS-F-001, VPS-F-002

4.2 Slot Management & Availability Display

Description: The system tracks slot occupancy in real time and displays availability to customers and administrators.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
VPS-F-004	The system shall display real-time slot availability (total, occupied, available) on the entry terminal and admin portal.	Functional	High	Customer / Admin	AC: Display updates within 5 seconds of any slot status change. Test: TC-SLOT-01	Requires DB polling or event-driven updates
VPS-F-005	The system shall support slot categorisation (e.g., regular, disabled, EV charging).	Functional	Medium	Operations	AC: Slot types configurable by admin; correct type assigned during booking. Test: TC-SLOT-02	Admin configuration required
VPS-F-006	The system shall prevent entry when all slots are occupied and display a "Parking Full" notice.	Functional	High	Operations / Customer	AC: Barrier remains closed and "Full" message displayed when occupancy = 100%. Test: TC-SLOT-03	Depends on VPS-F-002

4.3 Vehicle Exit & Fee Calculation

Description: On exit, the system validates the ticket, calculates the parking fee based on duration, processes payment, and opens the barrier.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
VPS-F-007	The system shall calculate the parking fee based on entry timestamp, exit timestamp, and the applicable rate schedule.	Functional	High	Business	AC: Fee = correct rate × duration; verified against test data. Test: TC-EXIT-01	Rate schedule configurable by admin
VPS-F-008	The system shall accept payment via cash, card, or digital wallet before opening the exit barrier.	Functional	High	Customer / Business	AC: Payment confirmed before barrier opens; failed payment prevents exit. Test: TC-EXIT-02	Requires payment gateway integration
VPS-F-009	The system shall print or send a digital receipt	Functional	Medium	Customer	AC: Receipt contains ticket ID, duration, amount	Depends on VPS-F-007, VPS-F-008

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
	upon successful payment at exit.				paid, and timestamp . Test: TC-EXIT-03	
VPS-F-010	The system shall mark the parking slot as available upon confirmed vehicle exit.	Functional	High	Operations	AC: Slot status reverts to "Available" within 5 seconds of exit confirmation. Test: TC-EXIT-04	Depends on VPS-F-002

4.4 Reservation

Description: Customers can pre-book a parking slot via the web portal or mobile interface.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
VPS-F-011	The system shall allow registered customers to reserve a parking slot in advance for a specified time window.	Functional	Medium	Customer	AC: Reservation confirmed with slot number and booking ID; slot held for 15 minutes past arrival time. Test: TC-RES-01	Requires customer account

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
VPS-F-012	The system shall automatically release a reserved slot if the customer does not arrive within the grace period.	Functional	Medium	Operations	AC: Slot released and marked "Available" after grace period expiry. Test: TC-RES-02	Depends on VPS-F-011

4.5 Admin & Reporting

Description: Administrators manage slots, rates, users, and access system reports.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
VPS-F-013	The system shall provide an admin dashboard displaying current occupancy, daily revenue, and transaction history.	Functional	High	Admin	AC: Dashboard loads within 3 seconds; data matches DB records. Test: TC-ADM-01	Admin login required
VPS-F-014	The system shall allow	Functional	High	Business / Admin	AC: Rate changes apply to	—

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
	administrators to configure parking rate schedules (hourly, flat rate, peak pricing).				all new transactions immediately after save. Test: TC-ADM-02	
VPS-F-015	The system shall generate exportable reports (CSV/PDF) for occupancy and revenue for selected date ranges.	Functional	Medium	Admin / Management	AC: Report exported in < 10 seconds; data matches on-screen dashboard values. Test: TC-ADM-03	Depends on VPS-F-013

5. Non-Functional Requirements

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
VPS-NF-001	Transaction processing (entry to ticket generation, exit to barrier open) shall complete within 5 seconds for 90% of	Performance	High	90th percentile ≤ 5s in load tests. Test: TC-PERF-01

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
	transactions under normal load.			
VPS-NF-002	The system shall maintain 99.9% availability per month; scheduled maintenance windows excluded.	Reliability	High	Uptime monitoring reports $\geq$ 99.9% per month. Test: Ops reports
VPS-NF-003	All payment card data must comply with PCI-DSS; card numbers must not be stored in plaintext.	Security / Compliance	High	PCI-DSS audit checklist pass. Test: TC-SEC-01
VPS-NF-004	The system shall retain transaction logs and audit events with timestamps for a minimum of 5 years.	Audit / Data Retention	High	Automated archival verified; retrieval tested. Test: TC-OPS-01
VPS-NF-005	Customer-facing terminals and web portal shall comply with WCAG 2.1 AA accessibility standards.	Usability / Accessibility	Medium	Accessibility audit pass. Test: TC-UX-01

5.1 Security

5.1.1 Security Objectives

1. **Protect cardholder and personal data** — Ensure all payment and personal data is encrypted in transit and at rest, preventing unauthorised disclosure or misuse.

2. **Prevent unauthorised system access** — Ensure only authenticated and authorised users (admins, attendants) can access management functions, and that all access is logged and auditable.

5.1.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance Criteria / Test Case Ref
VPS-SR-001	TLS 1.2+ shall be mandatory for all network communications between system components.	Security	High	TLS version verified via network scan. Test: TC-SEC-01
VPS-SR-002	All admin and attendant accounts shall be protected by password authentication with a minimum complexity policy (8+ characters, mixed case, number, symbol).	Security	High	Login with non-compliant password rejected. Test: TC-SEC-02
VPS-SR-003	The system shall implement Role-Based Access Control (RBAC); customers, attendants, and admins shall only access functions permitted for their role.	Security	High	Role boundary tests pass. Test: TC-SEC-03
VPS-SR-004	The system shall lock an account after 5	Security	High	Account locked after 5 failures; event logged.

Req ID	Requirement	Type	Priority	Acceptance Criteria / Test Case Ref
	consecutive failed login attempts and generate an audit event.			Test: TC-SEC-04
VPS-SR-005	License plate data and transaction records shall be encrypted at rest using AES-256 or equivalent.	Security	High	Encrypted storage verified in DB audit. Test: TC-SEC-05

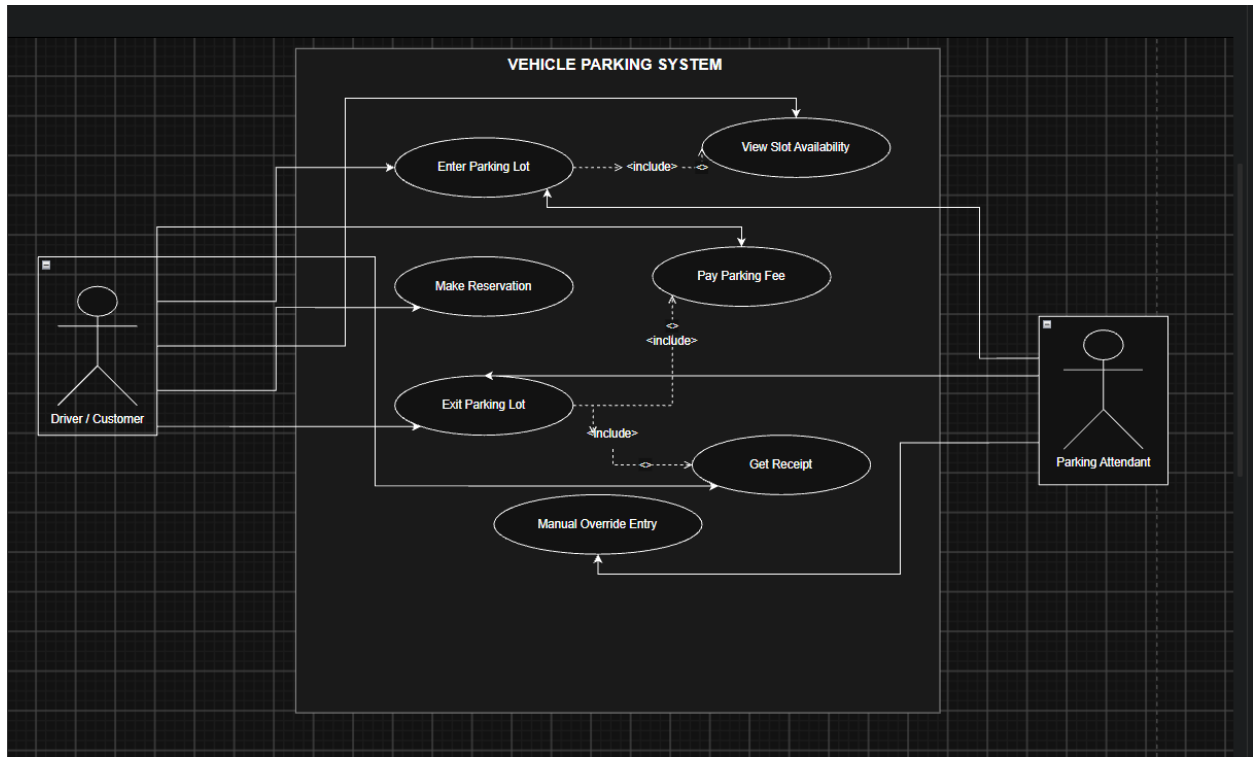
6. Quality Attributes & Acceptance Tests

Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR failures, and RTM shows all planned test cases passed.

Acceptance test suites: - Vehicle Entry & Exit flow (end-to-end) - Slot availability and capacity management - Fee calculation accuracy - Payment processing (card, cash, digital) - Admin dashboard and reporting - Performance under load (concurrent entries/exits) - Security and access control - Accessibility compliance

7. UML Use-Case Diagrams

7.1 Use-Case Diagram 1 — Customer Parking Flow



7.2 Use-Case Diagram 2 — Admin & Attendant Operations

Req ID	Requirement Short	Section Ref / Design Spec	Module	Test Case(s)	Status (N/P/A)	Comments
VPS-F-004	Real-time slot availability display	4.2 / DS-Slot-02	SlotManager / UI	TC-SLOT-01	N	
VPS-F-005	Slot categorisation	4.2 / DS-Slot-03	SlotManager	TC-SLOT-02	N	
VPS-F-006	Block entry when full	4.2 / DS-Slot-04	EntryModule	TC-SLOT-03	N	
VPS-F-007	Calculate parking fee	4.3 / DS-Fee-01	FeeCalculator	TC-EXIT-01	N	
VPS-F-008	Accept payment at exit	4.3 / DS-Pay-01	PaymentService	TC-EXIT-02	N	
VPS-F-009	Print/send receipt	4.3 / DS-Pay-02	ReceiptService	TC-EXIT-03	N	
VPS-F-010	Mark slot available on exit	4.3 / DS-Slot-05	SlotManager	TC-EXIT-04	N	
VPS-F-011	Advance slot reservation	4.4 / DS-Res-01	ReservationService	TC-RES-01	N	
VPS-F-012	Auto-release expired reservation	4.4 / DS-Res-02	ReservationService	TC-RES-02	N	
VPS-F-013	Admin occupancy dashboard	4.5 / DS-Adm-01	AdminPortal	TC-ADM-01	N	
VPS-F-014	Configure rate schedules	4.5 / DS-Adm-02	RateService	TC-ADM-02	N	
VPS-F-015	Export occupancy	4.5 / DS-Adm-03	ReportService	TC-ADM-03	N	

Req ID	Requirement Short	Section Ref / Design Spec	Module	Test Case(s)	Status (N/P/A)	Comments
	/revenue reports					
VPS-NF-001	Response time $\leq 5s$	5 / DS-Perf-01	All Modules	TC-PERF-01	N	
VPS-NF-002	99.9% availability	5 / DS-Rel-01	Infrastructure	Ops reports	N	
VPS-NF-003	PCI-DSS compliance	5 / DS-Sec-01	PaymentService	TC-SEC-01	N	
VPS-NF-004	5-year log retention	5 / DS-Ops-01	LoggingService	TC-OPS-01	N	
VPS-NF-005	WCAG 2.1 AA accessibility	5 / DS-UX-01	UI Layer	TC-UX-01	N	
VPS-SR-001	TLS 1.2+ for all connections	5.1.2 / DS-Sec-02	Network Layer	TC-SEC-01	N	
VPS-SR-002	Strong password policy	5.1.2 / DS-Sec-03	AuthService	TC-SEC-02	N	
VPS-SR-003	RBAC enforcement	5.1.2 / DS-Sec-04	AuthService	TC-SEC-03	N	
VPS-SR-004	Account lockout after 5 failures	5.1.2 / DS-Sec-05	AuthService	TC-SEC-04	N	
VPS-SR-005	AES-256 encryption at rest	5.1.2 / DS-Sec-06	DataLayer	TC-SEC-05	N	